Nishanka Das

Registration No. :A01-1112-117-003-2022
B.Sc.(Hons) in Computer Science
Ramakrishna Mission Vivekananda Centenary College
Website

+91-9432191900
nishanka.das.rkmvcc@gmail.com
nishanka2004@gmail.com
Github | LinkedIn
HackerRank

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Sc (Hons.)	Ramakrishna Mission Vivekananda Centenary College	8.75(Current)	2022-Present
Higher Secondary	WBBHSC Board	76.8%	2022
Secondary	WBBSE Board	89.7%	2020

Experience

- **Buddhanka Institute** *April. 2024 - Present*
  Faculty of Computer Science & Application Kolkata
  - Teaching Arduino based projects, MS EXCEL, Word, Power Point, HTML for class VI to X
  - Courses taught to class XI are Digital Logic, Python, Networking, Data Structure and Algorithm, Advanced Excel as per WBBHSC prescribed syllabus.
- **A.K Choudhury School Of Information Technology, University of Calcutta** *Oct 2023 - Present*
  Research Intern supervised by Prof. (Dr.) Amlan Chakrabarti, Director AKCSIT Kolkata
  - Currently learning CUDA Quantum for circuit implementation for various algorithms and machine learning algorithms
  - Comparing Classical Algorithms and Quantum Algorithms for optimized results with hybrid algorithms
- **Bharat Intern** *Aug. 2023 - Sept. 2023*
  Web Developer Intern Remote
  - Developed a portfolio website using HTML & CSS as assigned task.
  - Developed a calculating website using HTML, CSS & JS

Projects

- **Remote Sensing Imagery Classification using Hybrid Quantum Machine Learning** *Mar. 2024 - Present*
  AKCSIT, University of Calcutta, Kolkata supervised by Prof. (Dr.) Amlan Chakrabarti
  - Classification of water body from Multi-band Remote Sensing Imagery
  - Comparing Classical Support Vector Machine and Quantum Support Vector Classifier
- **Classical SVM and Hybrid QSVM Paradigms for Enhanced Credit Card Fraud Detection** *Oct. 2024 - Dec. 2024*
  AKCSIT, University of Calcutta, Kolkata supervised by Prof. (Dr.) Amlan Chakrabarti Github
  - Developed and implemented a comparative study of Classical Support Vector Machines (SVM) and Hybrid Quantum Machine Learning algorithms for detecting credit card fraud, achieving accuracies of 94.93% and 98.33%, respectively.
  - Designed and optimized a Quantum Support Vector Machine (QSVM) using 5-qubit angle encoding and advanced quantum kernel methods, showcasing superior performance in analyzing non-linear patterns within financial datasets.
- **Psychoinformatics with R** *June. 2022 - Jan. 2023*
  RKMVCC Rahara supervised by Prof. Joydeep Kumar Dam Drive
  - Analyzed the psychology of various people by taking survey and plotted the co-relation data using R programming language.
  - Used packages Psych, Corr plot

Skills

- **Programming**: Python, C, C++, Java, R
- **Operating Systems**: Windows, Linux
- **Non Technical**: Presentation, Linear Algebra, Graph Theory, Discrete Mathematics, LaTeX, Figma
- **Used Packages**: igraph(R), psych(R), ggplot(R), NumPy(Py), Matplotlib(Py), sklearn(Py), Pandas(Py), PyTorch(Py)
- **Quantum Programming**: Qiskit, CUDA Quantum

Key courses taken

- **Mathematics**: Linear Algebra, Basic Calculus, Discrete Maths, Abstract Algebra, Graph Theory
- **Computer Science**: C Programming , C++ Programming , Java Programming , Computer Networks, Operating System, Data Structures and Algo, Database Management System, Software Engineering
- **Quantum Computing**:
  - Basics of Quantum Information, IBM Quantum Education
  - Fundamentals of quantum algorithms, IBM Quantum Education

Certificates

- **Theory of Computation (NPTEL)**,
  Elite Certificate (65%) in Theory of Computation taought by Prof. Raghunath Tewari, IIT Kanpur *Oct. 2024*
- **Workshop on Introduction to Data Science and Machine Learning**,
  Attended "Workshop on Introduction to Data Science and Machine Learning" organized by the Department of Computer Science, RKMVCC, Rahara. Taought by Madhusudan Thakur, AI Researcher, American Express *Aug. 2024*
- **International Symposium on Advances in Informatics & Computing**,
  Participated in International Symposium organized by Department of Computer Science, RKMVCC, Rahara. Focused on Machine Leaning, Image Processing and Confidence Learning *Jan. 2024*

Positions of Responsibility

- **Volunteer at 27th ICPR 2024**,
  International Conference on Pattern Recolonization, Inida *Dec. 2024*
  - Designed keynote speakers' name plates
  - Volunteer at registration desk of ICPR
- **Nirdesh Question Setter and Designer**,
  Nirdesh (Inter College Tech Fest), RKMVC College, Rahara *April. 2024 - Sept. 2024*
  - Designed all the documents (Brochure, Hall Ticket, Schedule, Banner)
  - Prelims & Finals Question setter of Rahasyabhedi (Mystery Solving) Competition
- **Neuroverse Question Setter and Designer**,
  Neuroverse (Intra College Tech Fest), RKMVC College, Rahara *Feb. 2024 - Mar. 2024*
  - Designed all the documents (Brochure, Hall Ticket, Schedule, Banner)
  - Prelims Question setter of Codecraft(Coding competition) and prelims and final round question setter of Gray Matter (Mystery Solving Competition)

Achievements

- **Runner Up**, Poster presentation competition, Brainware University, Barasat *May. 2024*
- **Runner Up**, Poster presentation competition, GNIT, Sodhpur *Feb. 2024*
- **2nd Runner Up**, Secured **3nd rank** in intra college science quiz competition *Mar. 2024*

Extracurriculars

- **Student Volunteer**, National Service Scheme